

# James Wade

## NSF Graduate Research Fellow

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### EDUCATION

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**BACHELOR OF MECHANICAL ENGINEERING; MINORS IN COMP SCI & MATH** 2019 – 2020,  
Brigham Young University – Ira A. Fulton College of Engineering 2022 – 2026  
**GPA:** 4.0/4.0 (Rank #1 of 485 in class) | Emphasis in controls & robotic systems | **GRE:** Quant 170/170, Verbal 166/170  
**Russell M. Nelson Scholar:** Most prestigious scholarship at BYU awarded to <1% of students (four-year, full-ride)  
**Groups:** Mars Rover Team Lead, Rocketry Club, Spacecraft Club, Mechatronics Club, Global Engineering Outreach  
Note: I took a hiatus from my studies from 2020-2022 to serve a voluntary service mission in Mongolia

### RESEARCH, INTERNSHIPS & TECHNICAL LEADERSHIP

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**AUTONOMY INTERN** May 2025 – present

*Sandia National Labs, AutonomyNM Division – Albuquerque, NM*

- Designed RAM-Stack, a ROS-based autonomous multi-agent software stack for SLAM and navigation, integrating SLAM Toolbox, SwarmSLAM, Navigation2, and ROS 2 Control to enable coordinated mapping and path planning
- Leveraged theoretical understanding of Behavior Tree control flow (reactive sequences, fallbacks, and decorators) to diagnose and optimize Navigation2 logic
- Developing VLA-based behavioral policies for variable-formation drone swarms across simulation and hardware platforms (ongoing)

**SAFE CONTROL RESEARCH ASSISTANT** Aug 2024 – present

*Brigham Young University, MAGICC Lab – Provo, UT*

- Developing a discrete-time Control Barrier Function framework for bipedal robots using LiDAR projections in dynamic environments
- Using conformal prediction to provide probabilistic safety guarantees on the boundary of the safe region

**ROBOTICS RESEARCH ASSISTANT – Team Lead, Founding Member** Aug 2023 – present

*(One of Six National Finalists, NASA BIG Idea Challenge)*

*Brigham Young University, CMR Lab – Provo, UT*

- Founding member of the lab; established technical infrastructure and led an undergraduate team to develop a lunar-deployable soft robot
- Architected the redesign of a modular inflatable robot, implementing a custom multi-node radio network with error-checking protocols (manuscript in preparation for resubmission to *IEEE Transactions on Mechatronics*)
- Developing kinematic reconfiguration algorithms to maximize workspace and ensure fault tolerance under actuator failures (manuscript submitted to *IEEE RA-L*)
- Secured follow-on funding from the Utah NASA Space Grant Consortium through submitted proposal

**BYU MARS ROVER TEAM LEAD** August 2025 – May 2026

*Brigham Young University – Provo, UT*

- Elected by peers to lead 34-person team competing in 2025-2026 University Rover Challenge; manage technical subsystem leads across mechanical, electrical, and software divisions
- Oversaw system-level integration testing and served as primary technical resource for cross-subsystem debugging, leveraging comprehensive knowledge of the full robotics stack
- Architected migration from ROS 1 to ROS 2; developed training curriculum and led technical workshops to upskill team members in modern robotics software architecture

**EMBEDDED SYSTEMS INTERN** May 2024 – May 2025

*Lawrence Livermore National Laboratory (LLNL) – Livermore, CA*

- Engineered a custom, full-stack handheld controller (mechanical housing, circuit design, firmware) to teleoperate a fleet of autonomous marine rafts, replacing a cumbersome laptop-based system

- Programmed FreeRTOS task management and developed a custom LVGL-based control UI
- Delivered end-to-end mechanical, electrical, and software systems as the sole developer across the project lifecycle

## COMPLIANT MECHANISMS RESEARCH ASSISTANT

June 2023 – Aug 2023

*Mark Rober / Brigham Young University, CMR Lab – Provo, UT*

- Contributed to development of miniaturized compliant mechanisms for [The World's Smallest Nerf Gun](#) project (75M+ views on YouTube) across five size scales
- Conducted 100+ design iterations to identify failure modes and optimize mechanical performance
- Collaborated in weekly technical meetings to drive design refinements based on experimental data

## MICROFLUIDICS RESEARCH ASSISTANT

March 2023 – Aug 2023

*Brigham Young University, TEMP Lab – Provo, UT*

- Co-authored paper in *International Journal of Heat and Mass Transfer* on lab-on-a-chip thermal sensing using quantum dot fluorescence
- Conducted fluorescence characterization experiments and analyzed thermal response data to validate temperature measurement accuracy across operating conditions

## TEACHING EXPERIENCES

### ROBOTICS TEACHING ASSISTANT

Aug 2025 – December 2025

*Brigham Young University – Provo, UT*

- Selected as a TA for a 60-student graduate-level course on manipulators, kinematics, and dynamics
- Mentored students through theoretical derivations and ROS/Python implementations during weekly office hours

### SERVICE VOLUNTEER & ENGLISH TEACHER

June 2020 – June 2022

*The Church of Jesus Christ of Latter-day Saints – Ulaanbaatar, Mongolia*

- Completed two-year full-time service mission, teaching English 12+ hours/week in immigration offices, hospitals, and community centers while developing cross-cultural communication and adaptability skills
- Achieved conversational fluency in Mongolian

## PUBLICATIONS & PRESENTATIONS

### Publications

1. **James Wade**, Isaac Weaver, and Nathan Usevitch, “Fault-Tolerant, Rigidity-Preserving Control of Inflatable Truss Robots”, *IEEE Robotics and Automation Letters* [Submitted May 2026]
2. Mihai Stanciu, Isaac Weaver, Adam Rose, **James Wade**, Kaden Paxton, Chris Paul, Spencer Stowell, and Nathan Usevitch, “Modular Isoperimetric Soft Robotic Truss for Lunar Applications”, *IEEE Transactions on Mechatronics* [Submitted, February 2026; Preparing for Resubmission]
3. Derek Sanchez, Robert Macdonald, Brendan Mitchell, **James Wade**, McKay Wilkerson, Hunter Hinnen, Marshall Rawlins, Gregory P. Nordin, Adam T. Woolley, Troy R. Munro, “Advancing the applications of 3D printed microfluidics: Utilizing quantum dots to measure internal temperature”, *International Journal of Heat and Mass Transfer*, 2025.

### Conference Proceedings and Presentations: Podium

1. **James Wade**, “Fault-Tolerant, Rigidity-Preserving Control of Inflatable Truss Robots”, *Utah NASA Space Grant Consortium Symposium*, May 2025.
2. **James Wade**, Chris Paul, Mihai Stanciu, Spencer Stowell, Isaac Weaver, Adam Rose, Ashleigh Cerven, Annie O'Bryan, Brian Bodily, Logan Yang, and Nathan Usevitch, “Untethered and Modular Inflatable Robot for Lunar Applications”, *Finalists in the 2024 NASA BIG Idea Challenge — Inflatable Systems for Lunar Operations*. Technical paper presented at the NASA BIG Idea Symposium, November 2024.

3. **James Wade**, Chris Paul, “Constant-Pressure Untethered Soft Robotics: An Adaptable Solution to the Limitations of Soft Robots”, *Utah Conference on Undergraduate Research*, 2024. Announcement of our NASA grant.

#### **Conference Proceedings and Presentations: Poster**

1. **James Wade**, “ROS 2 Autonomy Stack for Multi-Agent SLAM and Navigation”, presented at *Sandia New Mexico SIP Intern Symposium*, August 2025.
2. **James Wade**, Chris Paul, “Untethered and Modular Inflatable Robot for Lunar Applications”, *Lunar Surface Innovation Consortium (LSIC)*, November 2024.

### **SCHOLARSHIPS & GRANTS**

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- 2026 **NSF Graduate Research Fellowship**, \$137,000 in research funding over three years
- 2025 **Utah NASA Space Grant Consortium Fellowship**, \$2,000 research stipend for work on multiply-redundant soft robots in space environments; received one of 17 available awards at BYU (14 graduates, 3 undergraduates)
- 2024 **NASA BIG Idea Competition Funding**, secured \$140,000 as head grant writer and team lead; one of six national finalists in 2024-2025 intercollegiate competition
- 2019 **President Russell M. Nelson Scholar**, 4-year full-ride scholarship for 150% of tuition, awarded to fewer than 1% of the student body (the highest scholarship available at BYU)

### **HONORS & AWARDS**

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- 2026 **Awarded the NSF Graduate Research Fellowship**
- 2025 **Ranked 1st in Mechanical Engineering Class of 485**
- 2025 **Dean’s List (all eight completed semesters)**, top 5% of BYU Engineering department
- 2023 **Phi Kappa Phi (honor society initiation)**, top 7.5% of BYU Engineering juniors

### **SELECT TECHNICAL PROJECTS**

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#### **Diffusion Models – From MNIST Generation to Robotic Control ([link](#))**

- Reimplemented Diffusion Policy (Chi et al.) and Diffuser (Janner et al.) from scratch, building a four-stage progression from DDPM/DDIM image generation to action-diffusion robotic control and classifier-guided trajectory planning

#### **Trebuchet Numerical Simulator – Graduate Dynamics Class Project ([link](#))**

- Derived unified equations of motion using Euler-Lagrange formulation with Lagrangian multipliers to compute constraint forces across three contact modes: ground sliding, pendulum swing, and ballistic flight
- Implemented a hybrid dynamics simulator with event-triggered mode transitions and constraint relaxation to extract normal/friction forces from variational mechanics

### **RELEVANT COURSEWORK & SKILLS**

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**Robotics & Control:** ROS 2 (Nav2, SLAM Toolbox, ROS 2 Control, MoveIt 2), ROS, Control Theory (PID, State-Space, Observers, MPC, LQR), State Estimation, Mechatronics, Dynamic System Modeling, Kinematics, Deep Learning

**Mechanical Design & Analysis:** SolidWorks (Certified Professional, Sheet Metal), Ansys FEA, Mechanical System Design, Thermodynamics, GD&T, 3D Printing

**Electrical & Embedded Systems:** Arduino, ESP32, STM32, FreeRTOS, Verilog, PCB Design

**Programming:** C/C++, Python, PyTorch, Rust, MATLAB, Julia, Qt, Git, LaTeX, Java